

SEMM Primer - Linear Algebra

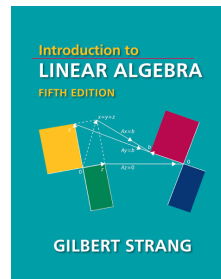
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Reference: Strang, 2016



① Linearity

Definitions

Sources and implications

Identifying and expressing linearity

② Systems of Linear Equations

③ Eigenvalue Problems

Linearity: Definitions

A linear function $\mathcal{A}(\boldsymbol{x})$ preserves addition and scalar multiplication.

$$\mathcal{A}(c\boldsymbol{x}) = c\mathcal{A}(\boldsymbol{x})$$

$$\mathcal{A}(\boldsymbol{x}_1 + \boldsymbol{x}_2) = \mathcal{A}(\boldsymbol{x}_1) + \mathcal{A}(\boldsymbol{x}_2)$$

Linearity: Definitions

A linear function $\mathcal{A}$ can be expressed in matrix notation.

$$\mathcal{A}(\boldsymbol{x}) = \mathbf{A}\boldsymbol{x}$$

Linearity: Sources and implications

Sources:

Kinematics **B** and Material **C**

Implications:

Stability, indeterminacy

Invertible stiffness

Modal properties

Linearity: Identifying and expressing linearity

$\mathcal{A}$ can be expressed in matrix notation. Example:

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(\mathbf{x}) = \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_1 + x_2 \end{bmatrix}$$

Linearity: Identifying and expressing linearity

$\mathcal{A}$ can be expressed in matrix notation. Example:

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(\mathbf{x}) = \underbrace{\begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_1 + x_2 \end{bmatrix}$$

quiz!



Linearity: Identifying and expressing linearity

$\mathcal{A}$ can be expressed in matrix notation. Example:

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(\mathbf{x}) = \underbrace{\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_1 + x_2 \end{bmatrix}$$

Linearity: Identifying and expressing linearity

Verify the linearity of $\mathcal{A}$.

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(c\mathbf{u}) = ?$$

$$c\mathcal{A}(\mathbf{u}) = ?$$

Are $\mathcal{A}(c\mathbf{u})$ and $c\mathcal{A}(\mathbf{u})$ equivalent?

If so, what is an equivalent expression for $\mathcal{A}(c\mathbf{u})$ and $c\mathcal{A}(\mathbf{u})$?



Linearity: Identifying and expressing linearity

Verify the linearity of $\mathcal{A}$.

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(c\mathbf{u}) = (2cu_1, cu_1 + cu_2)$$

$$c\mathcal{A}(\mathbf{u}) = c(2u_1, u_1 + u_2)$$

Yes. Both equal $(2cu_1, cu_1 + cu_2)$.

Linearity: Identifying and expressing linearity

Verify the linearity of $\mathcal{A}$.

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(\mathbf{u} + \mathbf{v}) = ?$$

$$\mathcal{A}(\mathbf{u}) + \mathcal{A}(\mathbf{v}) = ?$$

Are $\mathcal{A}(\mathbf{u} + \mathbf{v})$ and $\mathcal{A}(\mathbf{u}) + \mathcal{A}(\mathbf{v})$ equivalent?

If so, what is an equivalent expression for $\mathcal{A}(\mathbf{u} + \mathbf{v})$ and $\mathcal{A}(\mathbf{u}) + \mathcal{A}(\mathbf{v})$?



Linearity: Identifying and expressing linearity

Verify the linearity of $\mathcal{A}$.

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathcal{A}(\mathbf{u} + \mathbf{v}) = (2(u_1 + v_1), (u_1 + v_1) + (u_2 + v_2))$$

$$\mathcal{A}(\mathbf{u}) + \mathcal{A}(\mathbf{v}) = (2u_1, u_1 + u_2) + (2v_1, v_1 + v_2)$$

Yes. Both equal $(2u_1 + 2v_1, u_1 + u_2 + v_1 + v_2)$.

Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (\sin x_1, x_1 + x_2)$$

$$\mathbf{A} = ?$$



Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (\sin x_1, x_1 + x_2)$$

$\mathcal{A}$ is **not** a linear function: $c \sin x_1 \neq \sin cx_1$.

Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (2, x_1 + x_2)$$

$$\mathbf{A} = ?$$



Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (2, x_1 + x_2)$$

$\mathcal{A}$ is **not** a linear function: $c2 \neq 2$.

Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (0, 0)$$

$$\mathbf{A} = ?$$



Linearity: Identifying and expressing linearity

$\mathbf{x} \in \mathbb{R}^2$. Is $\mathcal{A}$ a linear function? If so, express the function as a matrix $\mathbf{A}$.

$$\mathcal{A}(\mathbf{x}) = (0, 0)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Linearity: Identifying and expressing linearity

A linear function $\mathcal{A}(\mathbf{x})$ preserves addition and scalar multiplication.

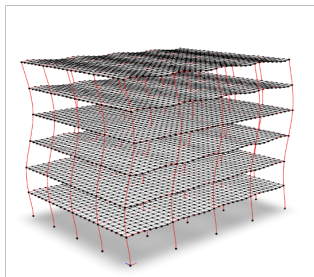
$$\mathcal{A}(c\mathbf{x}) = c\mathcal{A}(\mathbf{x})$$

$$\mathcal{A}(\mathbf{x}_1 + \mathbf{x}_2) = \mathcal{A}(\mathbf{x}_1) + \mathcal{A}(\mathbf{x}_2)$$

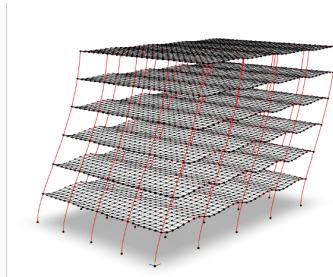
Use this fact when you need a systematic method to verify the linearity of a function.

Linearity

$$u_1 = \mathbf{K}^{-1} f_1$$



$$u_2 = \mathbf{K}^{-1} f_2$$



Building modeled by Goktug Tufekci in xara; rendered in veux

① Linearity

② Systems of Linear Equations

Matrix expression

Number of solutions

Static stability and determinacy

③ Eigenvalue Problems

Systems of Linear Equations: Matrix expression

$$b = Ax$$

Systems of Linear Equations: Matrix expression

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\mathcal{A}(\mathbf{x}) = (2x_1, x_1 + x_2)$$

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Systems of Linear Equations: Matrix expression

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$(b_1, b_2) = (2x_1, x_1 + x_2)$$

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Systems of Linear Equations: Number of solutions

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\mathbf{x} \in \mathbb{R}^n$$

$\text{rank}(\mathbf{A}|\mathbf{b}) = n \rightarrow$ one unique solution

$\text{rank}(\mathbf{A}|\mathbf{b}) < n \rightarrow$ infinite solutions

$\text{rank}(\mathbf{A}|\mathbf{b}) > n \rightarrow$ no solution

Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = ?$$



Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = 2$$

Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank}\left(\begin{bmatrix} 1 & 0.8 & 2 \\ 0.5 & 0.3 & 1 \end{bmatrix}\right) = ?$$



Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank}\left(\begin{bmatrix} 1 & 0.8 & 2 \\ 0.5 & 0.3 & 1 \end{bmatrix}\right) = 2$$

Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

How many solutions exist for $\mathbf{b} = \mathbf{A}\mathbf{x}$?



Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\text{rank}(\mathbf{A}|\mathbf{b}) = n = 2 \rightarrow \text{one solution.}$

Systems of Linear Equations: Number of solutions

Example 1

$$b = Ax$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Bonus! $x = \begin{bmatrix} ? \\ ? \end{bmatrix}$



Systems of Linear Equations: Number of solutions

Example 1

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Bonus! $\mathbf{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = ?$$



Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = 2$$

Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank}\left(\begin{bmatrix} 1 & 0.8 & 2 \\ 0.5 & 0.4 & 1 \end{bmatrix}\right) = ?$$



Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank}\left(\begin{bmatrix} 1 & 0.8 & 2 \\ 0.5 & 0.4 & 1 \end{bmatrix}\right) = 1$$

Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

How many solutions exist for $\mathbf{b} = \mathbf{A}\mathbf{x}$?



Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$(\text{rank}(\mathbf{A}|\mathbf{b}) = 1) < (n = 2) \rightarrow$ infinite solutions.

Systems of Linear Equations: Number of solutions

Example 2

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} c \\ \frac{5}{4}(2 - c) \end{bmatrix}$$

Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = ?$$



Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} \in \mathbb{R}^n. \quad n = 2$$

Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank} \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) = ?$$



Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{rank}(\mathbf{A}|\mathbf{b}) = \text{rank} \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) = 3$$

Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

How many solutions exist for $\mathbf{b} = \mathbf{A}\mathbf{x}$?



Systems of Linear Equations: Number of solutions

Example 3

$$\mathbf{b} = \mathbf{A}\mathbf{x}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(\text{rank}(\mathbf{A}|\mathbf{b}) = 3) > (n = 2) \rightarrow \text{no solution.}$$

Systems of Linear Equations: Number of solutions

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ one solution, "square"}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.8 \\ 0.5 & 0.4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ infinite solutions, "fat"}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ no solution, "tall"}$$

Systems of Linear Equations: Static stability and determinacy

$$\mathbf{p} = \mathbf{B}\mathbf{q}$$

$\mathbf{p}$: external forces; $\mathbf{B}$: equilibrium matrix; $\mathbf{q}$: internal forces

one solution, **square**: stable and determinate

infinite solutions, **“fat”**: indeterminate

no solution, **“tall”**: unstable

Reference: Pellegrino and Calladine, 1986

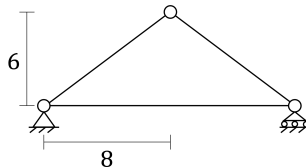
Systems of Linear Equations: Static stability and determinacy

Example 1

$$\mathbf{p} = \mathbf{B}\mathbf{q}$$

B:

- ☐ tall
- ☐ fat
- ☐ square



Systems of Linear Equations: Static stability and determinacy

Example 1

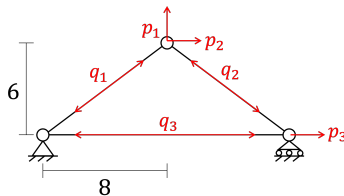
$$p = Bq$$

B:

- ☐ tall
- ☐ fat
- ☒ square

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.6 & 0 \\ 0.8 & -0.8 & 0 \\ 0 & 0.8 & 1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

unique solution: stable and statically determinate



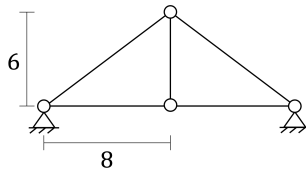
Systems of Linear Equations: Static stability and determinacy

Example 2

$$p = Bq$$

B:

- ☐ tall
- ☐ fat
- ☐ square



Systems of Linear Equations: Static stability and determinacy

Example 2

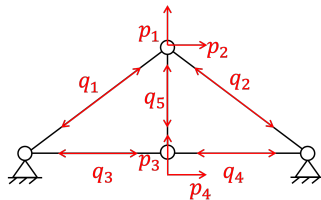
$$\mathbf{p} = \mathbf{B}\mathbf{q}$$

B:

- ☐ tall
- ☒ fat
- ☐ square

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.6 & 0 & 0 & 1 \\ 0.8 & -0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{bmatrix}$$

infinite solutions: statically indeterminate



① Linearity

② Systems of Linear Equations

③ Eigenvalue Problems

Motivation

Decoupling dynamic modes of vibration

Eigenvalue Problems: Motivation

Why?

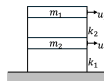
To decouple multiple degrees of freedom

Applications:

Principal stresses



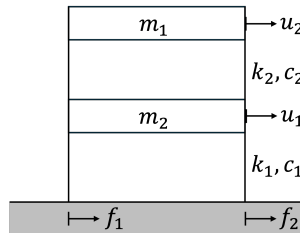
Dynamic modes of vibration



Eigenvalue Problems: Decoupling dynamic modes of vibration

The full system:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t)$$

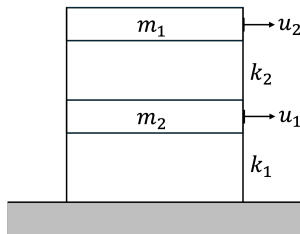


$$\mathbf{u} \in \mathbb{R}^n. \text{ e.g., } n = 2$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

The undamped system ($\mathbf{C} = \mathbf{0}$):

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{0}$$



$$\mathbf{u} \in \mathbb{R}^n. \text{ e.g., } n = 2$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

We know, or we'll see in the ODE primer, how to solve the scalar version of this equation: $m\ddot{u} + c\dot{u} + ku = 0$. But how to solve this vector version, where each degree of freedom, $u_1, u_2, \dots, u_n$ is coupled with the others?

Eigenvalue Problems: Decoupling dynamic modes of vibration

The system can be ***uncoupled*** into n natural modes of vibration, q_i .

$$\mathbf{u}(t) = \sum_i^n \phi_i q_i(t)$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

The system can be ***uncoupled*** into n natural modes of vibration, q_i .

$$\mathbf{u}(t) = \sum_i^n \phi_i q_i(t)$$

With the matrix Φ , we extract n decoupled scalar differential equations:

$$\mathbf{u}(t) = \Phi \mathbf{q}(t); \quad \ddot{q}_i + \lambda_{ci} \dot{q}_i + \lambda_{ki} q_i = 0.$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

The system can be ***uncoupled*** into n natural modes of vibration, q_i .

$$\mathbf{u}(t) = \sum_i^n \phi_i q_i(t)$$

With the matrix Φ , we extract n decoupled scalar differential equations:

$$\mathbf{u}(t) = \Phi \mathbf{q}(t); \quad \ddot{q}_i + \lambda_{ci} \dot{q}_i + \lambda_{ki} q_i = 0.$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

$$\mathbf{u}(t) = \Phi \mathbf{q}(t); \quad \ddot{q}_i + \lambda_{ci} \dot{q}_i + \lambda_{ki} q_i = 0.$$

Φ contains the eigenvectors of the “*generalized eigenvalue problem*” described by $\mathbf{M}$ and $\mathbf{K}$.

Eigenvalue Problems: Decoupling dynamic modes of vibration

The generalized eigenvalue problem:

$$(\mathbf{K} - \lambda \mathbf{M}) \phi = 0; \quad \det(\mathbf{K} - \lambda \mathbf{M}) = 0$$

$$\mathbf{K} \phi = \lambda \mathbf{M} \phi$$

Eigenvalue Problems: Decoupling dynamic modes of vibration

The generalized eigenvalue problem:

$$(\mathbf{K} - \lambda \mathbf{M}) \phi = 0; \quad \det(\mathbf{K} - \lambda \mathbf{M}) = 0$$

$$\mathbf{K} \phi = \lambda \mathbf{M} \phi$$

(Expressed as the classic eigenvalue problem:)

$$(\mathbf{M}^{-1} \mathbf{K} - \lambda \mathbf{1}) \phi = 0; \quad \det(\mathbf{M}^{-1} \mathbf{K} - \lambda \mathbf{1}) = 0$$

$$(\mathbf{M}^{-1} \mathbf{K}) \phi = \lambda \phi$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find λ :

$$\det(\mathbf{A} - \lambda \mathbf{1}) = 0$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find λ :

$$\det \left(\begin{bmatrix} 5 - \lambda & 0 \\ 8/3 & 3 - \lambda \end{bmatrix} \right) = 0$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find λ :

$$(5 - \lambda)(3 - \lambda) = 0$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find λ :

$$\lambda = 5, 3$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find ϕ :

$$(\mathbf{A} - \lambda \mathbf{1}) \phi = 0$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find ϕ :

$$\left(\begin{bmatrix} 5 - \lambda & 0 \\ 8/3 & 3 - \lambda \end{bmatrix} \right) \phi = 0$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find ϕ :

$$\lambda = 5; \quad \begin{bmatrix} 0 & 0 \\ 8/3 & -2 \end{bmatrix} \phi = 0; \quad \phi = \begin{bmatrix} c_1 \\ \frac{4}{3}c_1 \end{bmatrix}$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Find ϕ :

$$\lambda = 5; \quad \phi = \begin{bmatrix} c_1 \\ \frac{4}{3}c_1 \end{bmatrix}$$

$$\lambda = 3; \quad \phi = \begin{bmatrix} 0 \\ c_2 \end{bmatrix}$$

Eigenvalue Problems

Example

Determine the eigenvalues and eigenvectors of $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 8/3 & 3 \end{bmatrix}$$

Assemble Φ and Λ :

$$\mathbf{A} = \Phi \Lambda \Phi^{-1}$$

$$\Phi = \begin{bmatrix} 0.6 & 0 \\ 0.8 & 1 \end{bmatrix}; \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

References I

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